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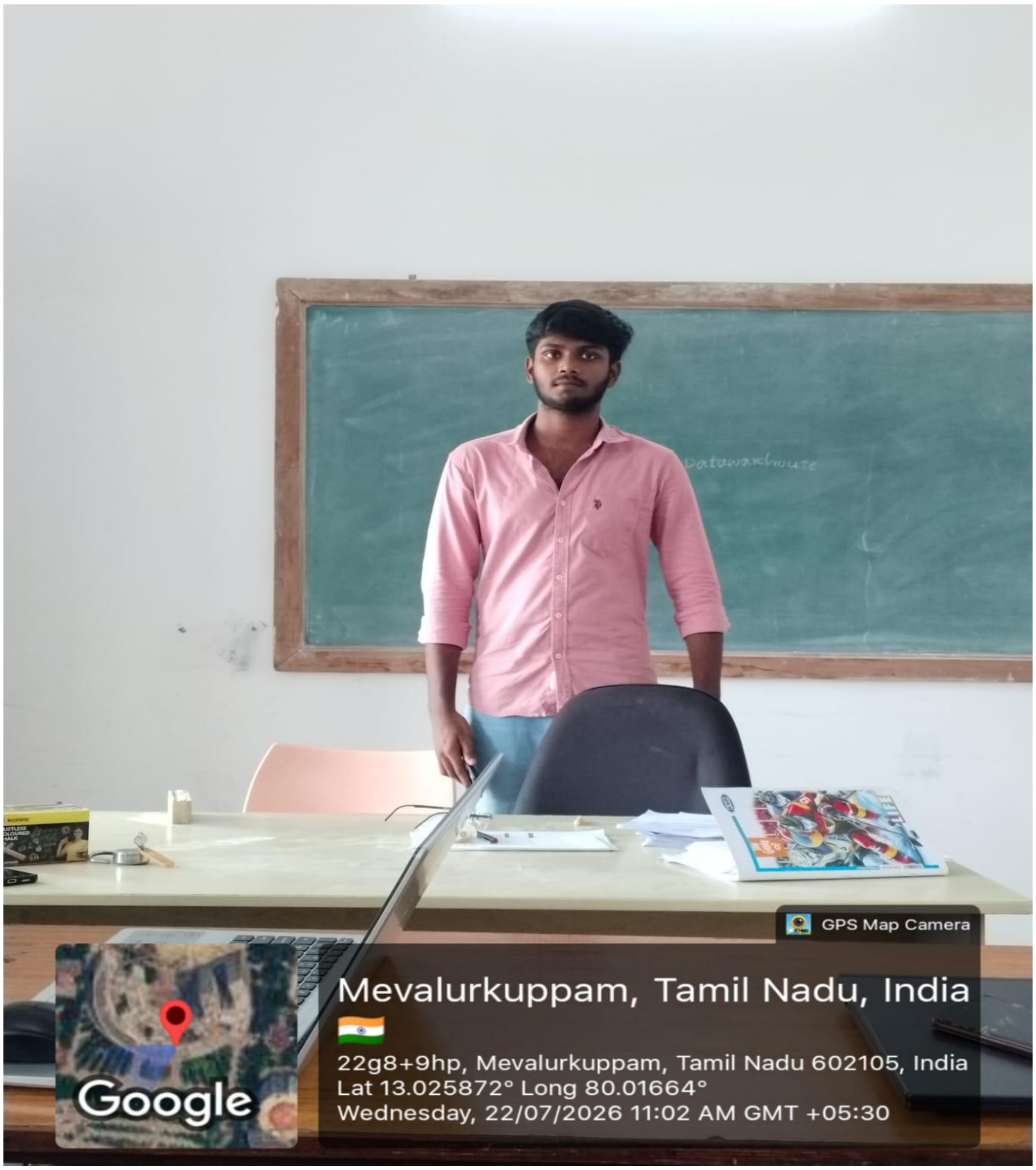

Data Warehousing and Data Mining are related concepts used in data analysis and decision-making, but they serve different purposes.

1. Data Warehousing

A data warehouse is a centralized repository that stores large amounts of data collected from different sources within an organization.

Purpose:

* To integrate, organize, and store historical data.
* To support reporting, analysis, and business intelligence.

Characteristics:

* Subject-oriented (organized around business subjects like sales, customers, etc.)
* Integrated (combines data from multiple sources)
* Time-variant (stores historical data)
* Non-volatile (data is stable and not frequently changed)

Example:
A retail company stores years of sales, inventory, and customer data in a data warehouse for analysis.

2. Data Mining

Data mining is the process of extracting useful patterns, trends, and knowledge from large datasets, often stored in a data warehouse.

Purpose:

* To discover hidden relationships and patterns.

* To support predictions and decision-making.

Techniques Used:

* Classification

* Clustering

* Association rule mining

* Regression

* Prediction

Example:

A supermarket analyzes customer purchase data to find that customers who buy bread often buy butter, helping improve product placement and marketing.

Relationship

Data warehousing provides the data foundation, while data mining extracts valuable insights from that data. In simple terms:

Difference Between Data Warehousing and Data Mining

Data Warehousing

Data Mining

- Stores and manages data

- Analyzes data to discover patterns
- Acts as a repository
- Acts as a knowledge discovery process
- Focuses on data collection and organization
- Focuses on extracting useful information
- Provides data for analysis
- Uses the stored data for analysis